BENCH BOX CHARGER — DRAFT SCHEMATIC (generated, unreviewed). Notes: charger/README.md.
USB-C PD sink (CH224A, 20 V) -> TI BQ24610 4S charger with input current limiting (DPM) and power path
(ACFET/RBFET + BATFET): the box runs from SYS, the charger gets what is left. Values per TI SLUS892D Figure 20.

USB-C input and PD sink

File: camry-charger-usb.kicad_sch

BQ24610 charger with input current limiting and power path

File: camry-charger-bq.kicad_sch

Connectors and mounting

File: camry-charger-connectors.kicad_sch

DRAFT generated by charger/kicad/gen_charger.py — not reviewed, not built

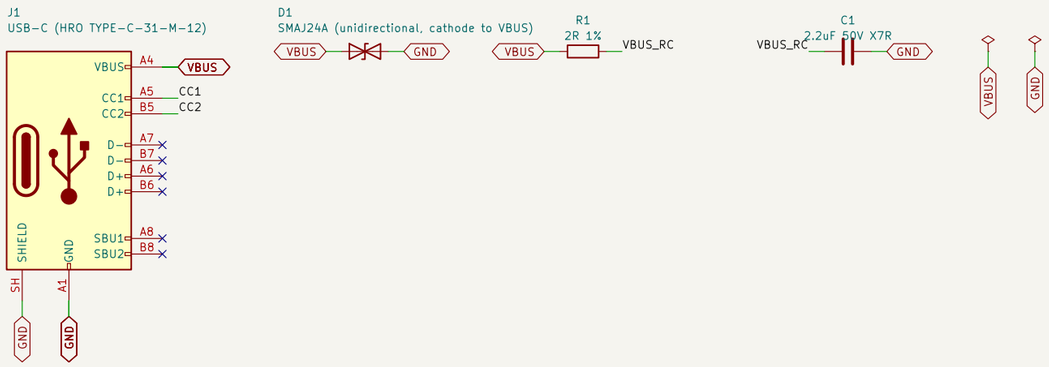
What If Garage — Drag Monster Camry

Sheet: /
File: camry-charger.kicad_sch

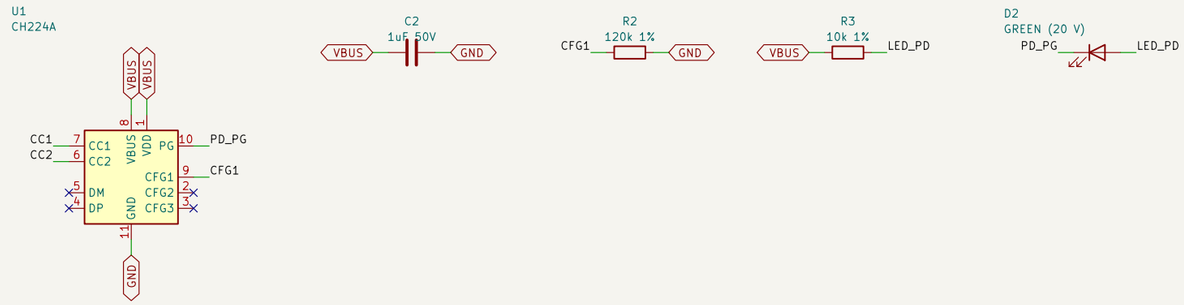
Title: Bench box charger

Size: A3	Date: 2026-11-03	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 1/4

USB-C RECEPTACLE (HRO TYPE-C-31-M-12, LCSC C165948, listed 5 A / 20 V). Power only: CC to the PD chip, D+/D- unused



PD SINK WCH CH224A (ESSOP-10, LCSC C42459160): asks the charger for 20 V. Above 3 A needs a 5 A e-marked cable



Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by charger/kicad/gen_charger.py — not reviewed, not built
What If Garage — Drag Monster Camry

Sheet: /USB-C input and PD sink/
File: camry-charger-usb.kicad_sch

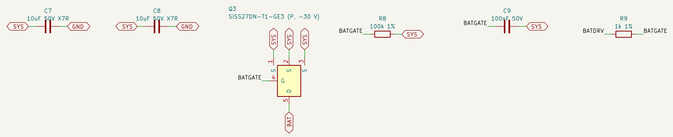
Title: Bench box charger — USB-C input and PD sink

Size: A3	Date: 2026-11-03	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 2/4

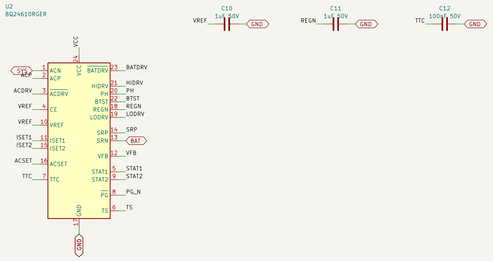
INPUT SWITCH: back-to-back P-MOSFETs (ACFET / RBFET) driven by ACDRV, input current sense RAC 10 mOhm



SYSTEM OUTPUT (to the rocker, then the 12 V converter and the USB hub) and BATTERY SWITCH (BATFET)



CHARGER IC TI BQ24610RGER (LCSC C19384). 600 kHz synchronous buck. Thermal pad to GND with vias (star point)



BUCK: high / low side N-MOSFETs, bootstrap, 6.8 uH, charge sense RSR 10 mOhm



SETTINGS (TI SLUS892D 9.3): VBAT = 2.1 V (1 + 681k/100k) = 16.40 V; ICHG = V(ISET1) / (20 x 10 mOhm) = 4.49 A; IDPM = V(ACSET) / (20 x 10 mOhm) = 2.92 A (jumper closed) or 4.38 A (jumper cut); ITERM = 0.30 A



BATTERY TEMPERATURE: 10k NTC (103AT) on the pack: 9.31k / 430k = 0-45 C charge window. No NTC: close JP to charge



STATUS LEDS (open drain to GND, fed from VBUS so they are dark on the pack)



THIRD BATTERY-SIDE CAPACITOR (review 2026-11-06): TI wants the output LC to resonate at 12-17 kHz (SLUS892D 10.2.1.2.6, 6.8 uH + 20 uF at 4 A). Two 10 uF 50 V X7R 1210 keep only ~6-7 uF each at 16 V (DC bias), ~17 kHz, at the edge; three give ~14 kHz. Last in the list so no other reference changes



Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by charger/kicad/gen_charger.py — not reviewed, not built

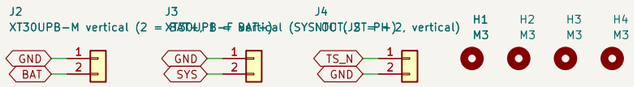
What If Garage — Drag Monster Camry

Sheet: /BQ24610 charger with input current limiting and power path/
File: camry-charger-bq.kicad_sch

Title: Bench box charger — BQ24610 charger with input current limiting and

Size: A1 Date: 2026-11-03	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6	Id: 3/4

CONNECTORS, all vertical (the board lies flat in the bay; plugs go in from above): battery (via the JBD BMS and the 10 A pack fuse), system out (to the rocker), pack NTC



Every pin is connected by net labels; KiCad ERC + check.py verify connectivity
DRAFT generated by charger/kicad/gen_charger.py — not reviewed, not built

What If Garage — Drag Monster Camry

Sheet: /Connectors and mounting/
File: camry-charger-connectors.kicad_sch

Title: Bench box charger — Connectors and mounting

Size: A3	Date: 2026-11-03	**Rev: draft 0.1**
KiCad E.D.A. 10.0.6		Id: 4/4